

I build AI systems that keep working after the demo.

Computer vision, vision-language models and agentic software, engineered for accuracy, reliability and modest hardware. I take systems from idea to working product: models, backends, interfaces and deployment.

5,000+**automated tests**

on a real-time video analytics platform

< 20 ms**per frame**

real-time 3D mapping

₹2,00,000**scholarship**

_VOIS Innovation Marathon 2.0 winner

7B**vision-language model**

fine-tuned and run on a single GPU

O+**both capstone projects**

B.Tech CSE, GLS University

8.4+**SGPA, last three semesters**

B.Tech CSE, GLS University

WHAT I BUILD

Computer vision

Detection, segmentation, tracking and OCR, evaluated on held-out data.

3D perception and robotics

Point-cloud perception, real-time mapping and robotics-middleware integration.

Operator consoles

React and Next.js dashboards with live video, 3D and map views.

Vision-language and LLM systems

Fine-tuned models, tool-using assistants and retrieval, with evidence and confidence on every answer.

Production AI backends

FastAPI services, PostgreSQL and Redis, Docker, ONNX and TensorRT optimisation.

Quality engineering

Automated tests, property-based testing and multi-hour soak runs.

THE STANDARD I BUILD TO

Quality

Held-out test sets, not cherry-picked demos. Results reported as measured, with the limits stated.

Scalability

Pipelines sized for the hardware you have: real-time systems that run on a single GPU.

Reliability

Thousands of automated tests and multi-hour soak runs before anything is called done.

Impact

Built for India's real problems: smarter cities, sustainability, national growth.

FROM FIRST CALL TO HANDOVER

1. Scope

A short call and a written scope with acceptance tests, so “done” is defined before work starts.

2. Build

Weekly demos on real data, not slides.

3. Verify

Held-out evaluation, automated tests and soak runs. You get the numbers, including the limits.

4. Hand over

Documented, tested code and deployment notes. You own it.

5. Support

Clear handover notes, and questions answered after delivery.

WHAT YOU CAN EXPECT

- Replies within one working day
- Trains and tests on a local 32 GB RTX 5090 workstation
- Works in India Standard Time
- AI coding agents in the workflow; every line reviewed and tested

SELECTED WORK

ReUpyog: A confidential AI project

Winner of the ₹2,00,000 _VOIS Innovation Marathon 2.0. SSIP seed grant, Government of Gujarat. Capstone Project I and II at GLS University, both graded O+.

Real-time video analytics platform

A GPU-accelerated platform with a full-stack operator console, engineered to production standards: proven under multi-hour soak tests and 5,000+ automated tests.

Real-time 3D perception & mapping system

Segmentation network trained from scratch; GPU-accelerated mapping under 20 ms per frame; 3D web console.

Agentic vision-language assistant

A fine-tuned 7B vision-language assistant on a single GPU that routes questions to specialist vision tools and backs every answer with evidence and a confidence score.

Technology-intelligence & forecasting platform

Smart India Hackathon 2025 finalist. Automates the collection, analysis and long-range forecasting of strategic-technology trends, with an offline GPU-powered AI engine, a secured API and an executive dashboard with one-click report export.

Details confidential, available to serious enquiries on request.

RECOGNITION

- Winner, _VOIS Innovation Marathon 2.0 · Feb 2026
- SSIP seed grant, Government of Gujarat · Mar 2026
- Finalist, Smart India Hackathon 2025 · 2025

CERTIFIED

- OCI Generative AI Professional
- OCI Data Science Professional
- OCI AI Foundations Associate

EDUCATION

- B.Tech, Computer Science & Engineering
- GLS University · CGPA 7.95 / 10